Individual Contributions to the Group Project

Group: 1

Teammate	Contributions to Team	Commendations	Percentage of Contribution
Eu Jin Yang	Project leadership and coordination across all development phases Full-stack development including Android application and backend services Research and documentation of technical feasibility and system constraints Developed application settings, preferences, and custom NFC reader implementation AuthViewModel implementation for authentication state management HCE service development for NFC card emulation Team collaboration & techinical support	Eu Jin Yang provided exceptional project leadership and full-stack development. He seamlessly coordinated our tasks and built robust features for both the front-end and back-end and implemented critical security mechanisms including hardware binding and HCE service. His technical expertise and clear direction were instrumental in navigating the project's pivot from direct card emulation to custom reader architecture. – Teoh Yi Shan Eu Jin Yang showed excellent leadership and strong full-stack development skills. He coordinated the team effectively and contributed significantly to both technical work and documentation, particularly in the areas of NFC implementation and security architecture. – Yong Ken Eu Jin Yang helped us by providing clear direction and effective project leadership,	25

		fostering a cohesive team environment. He also established a strong and well-defined project foundation through precise scoping, research, and critical decision-making when we encountered HCE limitations. – Ng Kean Chun	
Yong Ken	UML & Technical Modelling System Architecture documentation & design Gantt Chart Development Lead Collaboration & Quality Assurance Designed project poster Contributed to system design documentation ensuring clarity and consistency	Yong Ken did a spectacular job on developing Gantt Chart. He would ask for our help and keep on track of our progress to update the Gantt Chart, ensuring we stayed on schedule throughout the 24-week project timeline. – Eu Jin Yang Yong Ken did a tremendous job on UML modeling. He created incredibly clear and detailed diagrams that served as the essential blueprint for our entire development process, from registration flows to NFC authentication sequences. – Teoh Yi Shan Yong Ken did well for creating a coherent and well-thought-out system design and architecture diagram, which effectively	25

		guided technical development and ensured all components were properly integrated. – Ng Kean Chun	
Teoh Yi Shan	Client Communication Lead & Meeting Minutes Report Development authoring sections on Resources, Outcomes, Social Impact, and Sustainability Analysis (LSEP) Project Documentation & Administrative Support Hardware and Testing Coordination providing physical devices for real-world validation Contributed to system testing and quality assurance activities	Teoh Yi Shan handled client communication professionally and maintained clear meeting minutes. He contributed well-written report sections and supported the team's documentation tasks throughout the project lifecycle. – Yong Ken Teoh Yi Shan clearly articulated the project's vision, demonstrating a strong understanding of its broader impact and value, particularly in the areas of accessibility, inclusivity, and social considerations. – Ng Kean Chun Teoh Yi Shan's adept use of Android devices was an asset to the team. He readily provided his phone for us to test our system. He plays a crucial role in ensuring we can efficiently debug and deliver a fully functional product. This seemingly simple gesture was of great significance. – Eu Jin Yang	25
Ng Kean Chun	Developed the Image Capture and OCR module using Google ML Kit and CameraX	Ng Kean Chun single-handedly developed the core Image Capture and OCR	25

	Specified implementation plan and project background Designed application logo Hardware binding implementation for “One Device, One Account” security policy Contributed to requirement analysis and functional specifications Developed application profile tabs and QR code log activities Collaborated on testing and quality assurance	module from the ground up and designed the application’s logo and created a detailed execution plan. His work on OCR automated data entry from student cards, significantly enhancing user convenience. – Teoh Yi Shan Ng Kean Chun developed the image capture and OCR module, contributed to the implementation plan, and designed the app logo, adding both functional and visual value to the project. – Yong Ken It was a pleasure to work with Ng Kean Chun. He was always proactive, provided tremendous support, and reliably completed every task on time. His significant contributions to building the OCR system were the cornerstone of our project’s success. – Eu Jin Yang	

All group members agree to acknowledge the above contributions. (Yes/No)

*We understand that if we answered No to the above, we will be called for a viva with the Module Leader to mediate the dispute.

Signatures

Eu Jin Yang	
Yong Ken	
Teoh Yi Shan	
Ng Kean Chun	

Name: Eu Jin Yang

1. State Your Role

Team Leader & Full-Stack Developer / NFC & Security Lead

2. List Your Key Tasks

- Coordinated and hosted regular team meetings including weekly face-to-face sessions, daily stand-ups, and sprint planning / review meetings
- Assigned and delegated tasks based on group member's strengths to optimize workflow
- Conduct in-depth research on HCE technology and NFC communication limitations
- Led the technical pivot from direct campus reader emulation to custom NFC reader architecture
- Design the application's user interface (UI) in Figma
- Developed critical backend functions including AuthViewModel for authentication state management
- Implemented HCE service (HCECardService) for NFC card emulation using APDU protocol
- Developed SharedPreferences local storage for card UID persistence
- Co-wrote project documentation including system components, development approach, and requirements
- Co-created presentation slides and demonstration materials

3. Explain Your Impact

- Technical Architecture & Problem Solving
    - Led the critical decision to pivot from direct campus reader emulation to custom NFC reader architecture after discovering HCE limitations, ensuring project viability

- Enhanced Team Organization & Momentum
    - Implementing a structured Agile-Scrum workflow with 2-week sprint cycles, daily stand-ups, and sprint retrospectives kept the team on schedule and focused
    - Established clear communication protocols using WhatsApp for daily updates and GitHub for version control

- Improved System Quality
    - Led prototype simulation sessions to identify and resolve integration issues between Android app and ESP32 / PN532 hardware
    - Implemented lifecycle-aware synchronization ensuring local and cloud data consistency

- Elevated Presentation Standards
    - Oversaw the consolidation of teamwork into a cohesive and high-quality final presentation with live demonstrations

4. Mention Collaboration Efforts

- Fostered a collaborative environment by facilitating discussions on progress and future goals in every meeting

- Proactively gathered feedback from all members to inform decisions and ensure member's views were considered

- Provided consistent support to members on technical challenges (NFC implementation, Firebase integration, hardware debugging) and content-related tasks to ensure no one was blocked

- Coordinated closely with hardware team to ensure seamless integration between Android HCE service, ESP32 microcontroller and PN532 reader.

Name: Ng Kean Chun

1. State Your Role

Project Analyst & OCR Developer / Frontend Specialist

2. List Your Key Tasks

- Specified the background, problem statement, and objectives for the project proposal
- Developed and presented an implementation plan to guide team execution
- Designed and created the application logo to establish visual identity
- Developed the Image Capture and OCR module using Google ML Kit for text recognition
- Integrated CameraX library for camera hardware management and image analysis pipeline
- Developed hardware binding mechanism enforcing "One Device, One Account" security policy
- Implemented account termination with atomic batch deletions and complete data sanitization
- Developed application profile tabs and QR code log activities
- Specified functional requirements including core system features, user interactions, and data processing functions
- Collaborated on testing and quality assurance activities

3. Explain Your Impact

- Foundation & Vision
    - By defining the background and problem statement, I provided a clear and justified purpose for the project, ensuring all team efforts were aligned toward solving the real-world issue of access card dependency

- Technical Innovation
    - Developing the OCR module is a core technological innovation for our system. It automates data entry by scanning student cards, significantly enhancing user convenience and laying critical groundwork for the profile feature
    - Implementation using Google ML Kit's on-device text recognition ensures privacy and offline functionality

- Project Direction
    - The objectives and implementation plans created helped keep the team focused and on track throughout the development lifecycle

4. Mention Collaboration Efforts

- I actively participated in all meetings to discuss the project's direction and integrate my components with other's work.

- Worked with the backend and UI developers to ensure the OCR module could be properly integrated into the application's architecture.

- Engaged with the team to seek and provide feedback on further improvements, design choices and technical approaches.

Name: Teoh Yi Shan

1. State Your Role

Client Communication Lead & Meeting Secretary / Documentation Coordinator

2. List Your Key Tasks

- Client Communication Lead
  - Acted as the primary point of contact for client communications
  - Managed stakeholder expectations and provided regular project updates

- Meeting Minutes
  - Recorded and organized meeting minutes for all internal and client meetings
  - Maintained clear records of decisions, action items, and technical discussions

- Report Development
  - Authored significant sections of the project report including Introduction (Project Overview, Purpose and Objectives, Target Users, Scope and Limitations)
  - Contributed to Social Impact and Sustainability Analysis (LSEP) sections
  - Assisted in documentation of project management and communication strategies

- Hardware & Testing Coordination
  - Provide personal mobile phone for the development team to test the system's functionality and compatibility
  - Coordinated hardware testing activities ensuring real-world validation

3. Explain Your Impact

- Ensured Clear Communication
  - Maintained consistent and professional communication channels with clients and stakeholders, ensuring expectations were managed and feedback was properly incorporated.

- Maintained Project Organization
  - Detailed meeting minutes provided a reliable record of decisions and progress, enabling team members to stay aligned even when unable to attend meetings.

- Strengthened Project Reporting
  - Well-structured report sections contributed to a comprehensive final deliverable that clearly articulated the project's purpose, scope, and value proposition.

- Facilitated Practical Testing

- Provision of real Android devices enabled accurate testing of hardware-dependent features (OCR, NFC, HCE) that cannot be simulated effectively on emulators.

4. Mention Collaboration Efforts

- Actively attended and contributed to all scheduled team meetings including weekly sessions, sprint reviews, and retrospectives.
- Shared drafted report sections with teammates for feedback and integrated their suggestions to improve content quality.
- Coordinated closely with the technical team to understand system features for accurate documentation and client updates.
- Supported team members by taking on documentation tasks, allowing them to focus on their core development and research responsibilities.

Name: Yong Ken

1. State Your Role

System Design & Planning Lead / UML Modeling Specialist

2. List Your Key Tasks

- Created multiple UML diagrams such as use-case diagram and class diagram
- Led the development of the project Gantt chart, organizing sprint timelines across 24 weeks and 5 phases (Planning, Design, Core Development, Testing, Documentation)
- Contributed to system architecture documentation including architectural patterns, component interactions, and data flow mechanisms
- Assisted in quality assurance and code review activities
- Designed the project poster for presentation

3. Explain Your Impact

- Provided Technical Blueprint
    - My system design work gave the team a clear and structured blueprint, ensuring every member understood the technical flow and requirements. The UML diagrams served as essential references throughout development, particularly for understanding complex HCE communication sequences

- Ensured Project Timeliness
    - The Gantt chart planning ensured the group's work was well-organized, realistic, and evenly distributed across 24 weeks, reducing confusion about deadlines and responsibilities. Clear milestone tracking enabled the team to monitor progress and adjust priorities as needed

4. Mention Collaboration Efforts

- Coordinated closely with group members to ensure that designs accurately reflected implementation requirements and aligned with deadlines.

- Shared regular updates on design progress and clarified technical requirements when team members needed guidance.

- Maintained positive environment by actively seeking feedback on diagrams and documentation, incorporating suggestions for improvement.